

Lab 5 Use Case Diagrams

Supporting tool - Visual Paradigm | 2 hours 30 minutes

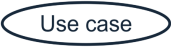
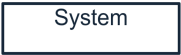
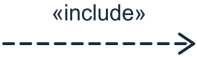
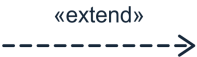
Learning objectives

- Identify actors, their goals and the system boundary.
- Draw and justify associations, include, extend and generalization.
- Review the model and describe a successful outcome ready for Lab 6.

Course learning outcome (CLO 2): Elicit, analyze, prioritize and document requirements for a software system.

Use Case Diagram

A use case diagram shows actor goals and participation; a DFD shows data transformations and storage.

Name	Symbol	Meaning and direction
Actor		An external role or system that interacts with the system.
Use case		A service or goal, named with a verb and an object.
System boundary		Names the system. Use cases go inside; actors stay outside.
Association		A solid line showing participation, not execution order.
Include		Dashed open arrow from the base to required supporting behaviour.
Extend		Dashed open arrow from optional behaviour to the base use case.
Generalization		Solid line with a hollow triangle pointing to the general actor or use case.

Part A QU Print Shop - Guided practice

Work through A.1 to A.4 with the instructor (35 minutes).

A.1 Recall the system and identify actors

Yellow highlighting marks the additions for Lab 5.

The QU Print Shop provides document printing services to students **and staff members**. A user uploads a document and selects the printing options: the number of copies, colour or black-and-white printing, and single- or double-sided printing.

Before accepting a print job, the Print Shop checks whether the user has sufficient credit. Credit balances are maintained by the Print Account System (PAS), a commercial package already operated by the university. The Print Shop does not maintain balances or collect money. If sufficient credit is available, the job is accepted and queued for printing; otherwise, the request is rejected and the user is informed.

Users may revise the printing options of a queued job. Before accepting either a new job or revised options, the Print Shop checks with PAS whether the user has sufficient credit for the resulting printing cost.

The Print Shop keeps a record of each accepted job, including who submitted it, the selected printing options, and its status: queued, printed or collected. Accepted, queued jobs are processed automatically. After printing a job, the Print Shop marks it as printed and sends charge advice to PAS, which processes the charge. The Print Shop also notifies the user that the printed document is ready for collection.

When collecting the printed document, the user submits a collection request, and the Print Shop records the job as collected. The user may also request a collection receipt, which the Print Shop issues once collection has been recorded.

Staff members can additionally request a summary of their own print jobs. This summary is produced from the Print Shop's job records and includes only the requesting staff member's jobs.

A.2 Identify actors and goals before drawing

Work through these five steps together. Use evidence from A.1 and state the result of each interaction before choosing a use case name.

Step 1 Establish the boundary

The Print Shop manages jobs, printing, notifications and collection records. PAS maintains credit balances and processes charges. Keep PAS outside the Print Shop boundary; sending charge advice is the Print Shop's responsibility.

Step 2 Identify who interacts with the system

Actor	Actor type	Interaction with the Print Shop
User	Primary	Represents the common role of students and staff: submits print requests, collects printed documents and receives notifications.
Student	Primary	Specializes User and inherits its interactions.
Staff member	Primary	Specializes User and inherits its interactions. Also requests a summary of their own print jobs.
PAS	Secondary	Provides credit-check results and receives charge advice. It is an external supporting system.

Step 3 Identify a purpose and a successful outcome

For each interaction, ask: **What is its purpose**, and **what will be true when it succeeds**? Name use cases with a verb and an object.

Use case ID: UC001. UC stands for Use Case; the three-digit number is a stable identifier, not an execution order. Keep IDs consistent across the inventory, diagram and later specifications.

Use case ID	Use case	Successful outcome
UC001	Submit print job	A job with sufficient credit is recorded and queued with its document and options.
UC007	Revise queued print job	Revised printing options are saved for the queued job after sufficient credit is confirmed.
UC002	Process print job	The job is printed and marked printed; charge advice and a ready notification are sent.
UC003	Collect printed document	Collection is recorded and the job is marked collected.
UC004	Issue collection receipt	A requested receipt is issued after collection is recorded.
UC005	Request own print-job summary	Only the requesting staff member's jobs appear in the summary.
UC006	Check sufficient credit	PAS confirms that available credit covers the resulting printing cost.

Step 4 Choose the right level of detail

- Uploading a document and selecting options are steps within UC001. Sending charge advice and notifying the user remain steps within UC002.
- UC002 runs automatically for accepted, queued jobs. User is associated with UC002 because they receive the ready notification, not because they start processing.
- UC006 is supporting behaviour shared by UC001 and UC007. Extracting it avoids describing the same credit check twice. Not every required step needs a separate oval.
- Submission ends with a queued job; processing ends with a printed job. Their order alone does not justify an include relationship.

Step 5 Select and justify relationships

Relationship	Print Shop example	Reason
include	UC001 and UC007 include UC006.	Both submission and revision require the same credit check. Each include arrow points to UC006.
extend	UC004 extends UC003.	Receipt condition: [receipt requested]. Extension point: collection recorded. Collection can finish without a receipt.
generalization	Student and Staff member specialize User.	Both inherit User interactions. Only Staff member participates in UC005.

A.3 Draw the use case diagram

- In Visual Paradigm, choose Diagram > New > Use Case Diagram.

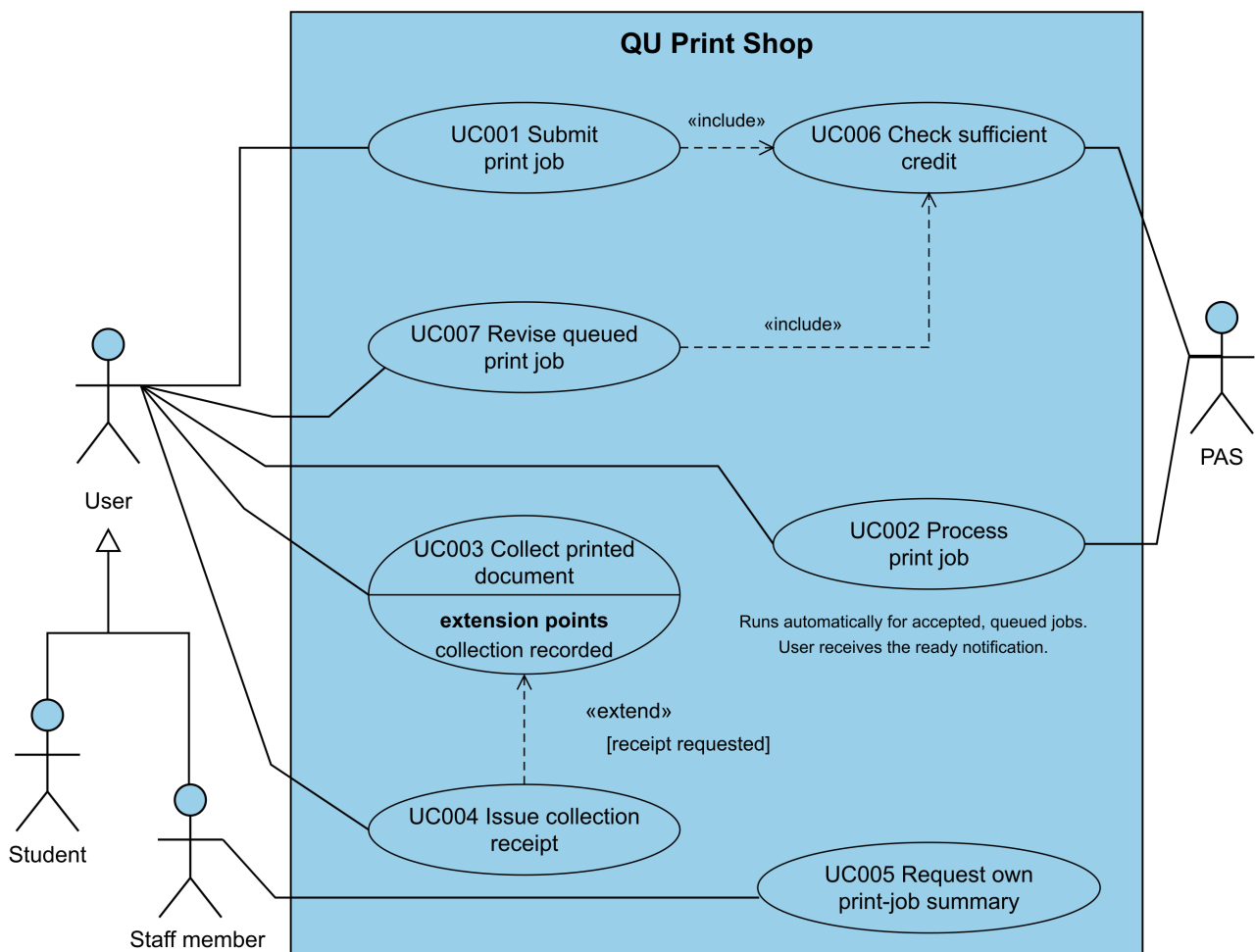


Figure 1 QU Print Shop Use Case Diagram

A.4 Review the diagram

- Why is User associated with automatic processing (UC002)? Can collection finish without a receipt?

Part B All Phones Shop

Individual assessment. Use the APS case from [Lab 3](#). You may discuss ideas, but create and explain your own model. Use earlier diagrams to check scope, not to copy a list of use cases.

B1 Define the system boundary and identify actors - 10 minutes

- Write a boundary statement in two or three sentences. List actor roles and support each with an interaction from the case.
- Explain individual and corporate clients, and how buying and selling roles relate to them.
- Keep PoB responsible for payments and financial accounts; APS handles registration, sales and invoices. Login is outside scope. Do not invent staff roles or external systems.

B2 Build the use case inventory - 25 minutes

Use verb–object names and IDs UC001, UC002 and so on. Start a separate sequence for the APS model. Complete this table, adding rows as needed.

Use case ID and name	Participating actors	Successful outcome	Case evidence

Check coverage of:

- Registration and removal; listings, image approval and withdrawal.
- Fixed-price purchases; bidding and auction completion.
- Invoicing, payment notifications and overdue sales.

For auction behaviour, distinguish placing a bid from the later result. Identify the trigger and participants; time passing does not itself justify a Clock actor. Record unresolved questions as assumptions for instructor review.

B3 Draw and refine the diagram - 35 minutes

- Create APS Use Cases in Visual Paradigm. Put use cases inside an All Phones Shop boundary and actors outside. Match the inventory's IDs and names.
- Save an initial draft, then refine it. Connect actors only where they participate. Keep business rules in notes; omit DFD stores and sequence arrows.
- Record one decision for each relationship type below. A justified decision to omit a type is acceptable; no number of connectors is required.

Type	Candidate elements	Use or omit and why
include		
extend		
generalization		

- Include: identify the required behaviour.
- Extend: state the condition and extension point; explain why the base can stand alone.
- Generalization: check inherited behaviour and permissions. Save the refined model.

B4 Review and improve the model - 20 minutes.

Exchange diagrams with a classmate and check:

- Boundary and actors: external roles and interactions are supported by the case.
- Coverage: meaningful goals cover the B2 areas.
- Relationships: types, arrow directions and inherited permissions are correct.
- Consistency: names and IDs match the inventory; connections are readable.

Record one strength, one specific improvement and the change you made. Resolve uncertain choices with the instructor and update both the diagram and inventory.

Submission requirements

15 minutes. Submit these three files to your-repo/05-use-cases/ using the individual repository workflow from Lab 4. Part A is not submitted.

File	Required contents
lab05-solution.docx	Name and student ID; B1 boundary and actors; B2 inventory; final diagram and B3 decisions; B4 review and revision; B5 outcome; assumptions.
aps-use-cases.png	Readable export of the final APS diagram.
aps-use-cases.vpp	Editable Visual Paradigm project containing both initial and final diagrams.

Commit and push all three files, then check that they open from the repository. Bring your final diagram and inventory to Lab 6.

Assessment criteria (10 points)

Criterion	Points
Correct boundary, supported actors and appropriate role permissions	2
Meaningful use cases, case coverage and evidence in the inventory	3
Correct relationships and justified decisions to use or omit them	2
Correct notation, readability and consistent names and IDs	1
Peer feedback acted on and a clear successful outcome for Lab 6	1
Complete submission in the required formats	1

Further reading

[Visual Paradigm use case diagram guide](#)

[OMG UML 2.5.1 Section 18 Use Cases](#)